

Linear Algebra - HW 2 Solutions

1) a)	In	Out	Equation
A:	300+500	$x_1 + x_2$	$x_1 + x_2 = 800$
B:	$x_2 + x_4$	$x_3 + 300$	$x_2 - x_3 + x_4 = 300$
C:	400+100	$x_4 + x_5$	$x_4 + x_5 = 500$
D:	$x_1 + x_5$	600	$x_1 + x_5 = 600$
All:	300+500 +100+400	300+ x_3 +600	$1300 = x_3 + 900$ $x_3 = 400$

System of equations:

$$\begin{cases} x_1 + x_2 = 800 \\ x_2 - x_3 + x_4 = 300 \\ x_4 + x_5 = 500 \\ x_1 + x_5 = 600 \\ x_3 = 400 \end{cases}$$

matrix:

$$\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ 1 & 1 & 0 & 0 & 0 & 800 \\ 0 & 1 & -1 & 1 & 0 & 300 \\ 0 & 0 & 0 & 1 & 1 & 500 \\ 1 & 0 & 0 & 0 & 1 & 600 \\ 0 & 0 & 1 & 0 & 0 & 400 \end{bmatrix}$$

b) solve the above matrix to answer this!

you should get:

$$\begin{aligned} x_1 &= 600 - x_5 \\ x_2 &= 200 + x_5 \\ x_3 &= 400 \\ x_4 &= 500 - x_5 \\ x_5 &= \text{free} \end{aligned}$$